CVT/PVT — Workflow & Study-Design Questions

Hi Utkarsh,

Before I start building, there are a few workflow and study-design decisions that need your or Dr. Poltavski's input. I've grouped them below; where I have a recommendation I've noted it — feel free to override.

Questions 3, 8, and 10 are the most blocking — they'll shape how I sequence the next development phase. No rush on the rest; happy to discuss in person if easier.

Hardware & Software

1. Smarteye's role. How is Smarteye being used alongside the Tobii — redundant eye-tracking, head tracking, or something else? Does it route through iMotions, or do we need to handle it separately?

2. Software versions on the lab machine. Could you let me know which versions of iMotions, B-Alert Live, Tobii Pro Lab, and Smarteye are installed?

iMotions Setup

3. iMotions study definition. Building the iMotions study (sensor list, recording settings) needs to happen once before any participant runs. Will you or Dr. Poltavski set that up, or should I get iMotions access to build it?

Recommendation: one shared study (e.g. `Vigilance_CVT_PVT`) with a new respondent added per participant.

Event Markers

4. Confirm the event list. I plan to mark: block start/end, each 6-minute period transition, stimulus onset, participant response, practice start/end, and the difficulty condition. Should we also mark stimulus offset, or is onset enough?

5. Stimulus marker granularity (CVT). For each trial, three options:

- (a) onset only
- (b) onset + offset as a paired range marker
- (c) onset, offset, and a separate response marker

6. Should marker labels reveal signal vs. non-signal? (e.g., `cvt_signal_onset` vs. `cvt_nonsignal_onset`.) Makes offline epoching trivial; only matters if someone is watching the iMotions timeline live during recording.

7. Timing-precision target. What alignment precision do you need between markers and the EEG / eye-tracking streams for the planned analyses (Engagement Index, Frontal Theta, fixation/saccade events)? Sub-millisecond is achievable on the same machine; if tighter is required I would add a hardware trigger.

Session Workflow

8. Who starts the iMotions recording?

- (a) The RA clicks Start in iMotions, then launches the task software. Task only sends markers.
- (b) The task software triggers iMotions to start recording automatically.

9. Recording continuity. iMotions starts/stops all sensors together — there's no public way to toggle EEG and eye-tracking independently. Should we use:

- (a) one continuous recording for the entire session (EEG + eye-tracking through practice, breaks, and both tasks), or
- (b) separate recordings for CVT and PVT?

Option (a) is simpler and preserves a continuous EEG baseline; (b) gives smaller, task-scoped files.

10. Calibration timing. I'm assuming this order: RA seats participant → runs Tobii calibration and B-Alert impedance in iMotions → launches the task software → task shows the EEG-baseline hold screen → recording proceeds. Does that match what you've been planning? Is re-calibration ever expected between blocks (e.g., if the participant shifts)?

11. Failure handling. If iMotions disconnects mid-block (rare but possible), should the task:

- (a) abort the block
- (b) continue, log a warning, finish the behavioral data

Thanks!

Jeff